AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# M269
# Algorithms, Data Structures and Computability

## Midterm Examination (MTA)
## Summer Semester 2025/2026

**Time Allowed: 2 Hours**

**Instructions:**

- Total Marks: 60
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted
- Be sure you write your name and ID on the External Answer booklet.

AOU EGYPT






CamScanner

## PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

**Questions 1: (10 marks)**

1. What is the time complexity of two consecutive loops that run from 0 to n?
   A. O(1) C. O(log n)
   B. O(n) D. $O(n^2)$
2. Which of the following functions represents quadratic growth?
   A. n C. $n^2$
   B. log n D. $2^n$
3. Which of the following sorting algorithms does not have a time complexity of $O(n^2)$?
   A. Insertion Sort C. Quick Sort
   B. Heap Sort D. Bubble Sort
4. What is the main disadvantage of Merge sort algorithm?
   A. Hard to implement C. Worst-case performance
   B. Requires a lot of memory D. Both Answer A and answer B
5. Which sorting algorithm is generally more efficient for large datasets?
   A. Bubble Sort C. Heap Sort
   B. Insertion Sort D. Selection Sort
6. Which of the following is considered the fastest algorithm in terms of complexity?
   A. O(n log n) C. $O(n^4)$
   B. $O(n^2)$ D. O(n)
7. What type of data structure are commonly used for last-in-last-out discipline
   A. Stack C. Linked list
   B. Tree D. Queue
8. Why is growth rate important in algorithm analysis?
   A. It determines the exact execution time of the program
   B. It determines the exact memory of the program
   C. It indicates how the execution time increases with input size
   D. Both answer A and answer C
9. Which of the following correctly identifies the ranking of complexities from most efficient to least efficient?
   A. $O(n\log n)<O(n)<O(n^2\log n)<O(n^2)$ C. $O(n)<O(n^2)<O(n\log n)<O(n^2\log n)$
   B. $O(n)<O(n\log n)<O(n^2)<O(n^2\log n)$ D. $O(n\log n)<O(n^2)<O(n)<O(2^n)$
10. In a minimum heap, what is the relationship between a parent node and its children?
    A. The parent node is always smaller than or equal to its children
    B. The parent node is always equal to its children
    C. The parent node is always larger than or equal to its children
    D. The parent node has no specific relationship to its children

## PART 2: ALL QUESTIONS ARE REQUIRED [30 Marks]

**Question 2: (15 marks)**

a) What is the complexity of the following tasks? (8 marks)
1) Arranging a collection of $n$books on a shelf in alphabetical order using the Quick-Sort method.
2) Processing the next passenger waiting in a queue of $n$ people at an airport check-in counter.
3) Locating the cheapest product in a linked list containing $n$ product records.
4) Retrieving the most recently added document from a stack containing $n$ documents.
5) Sorting a list of $n$ students who all have exactly the same grade using the Bubble-Sort method.
6) Merging two already sorted lines of students into a single sorted line.



Arab Open University
Arab Republic of Egypt

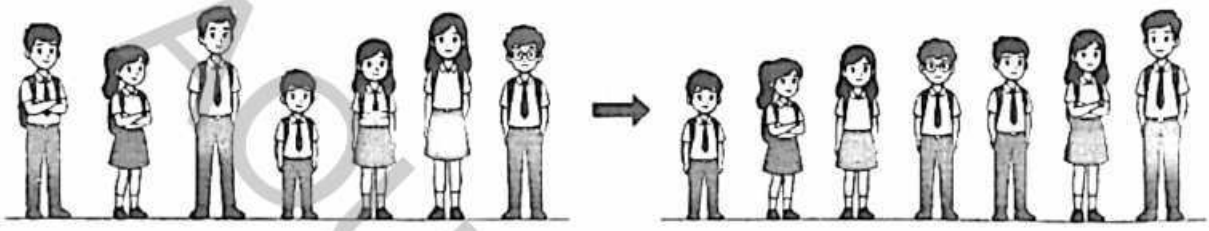
CamScanner

7) Looking up a student's record using their student ID in a well-organized database where the exact location is known
8) Finding the student standing at the front of a line.

b) For each of the following cases select the "best" data structure: (7 marks)
1) Managing a stack of trays in a cafeteria where the most recently placed tray is removed first.
2) A drawing application that allows users to reverse their most recent action.
3) Navigating forward and backward between web pages in a browser.
4) Organizing departments and sub-departments within a company organizational chart.
5) Vehicles passing through a toll booth in the order they arrive.
6) Maintaining a dynamic collection of student records where entries are frequently added, removed, or updated.
7) Managing train cars where cars can be attached or detached from either end efficiently.

## Question 3: (15 marks)

Given a group of *n* students with different heights standing in a line. The goal is to arrange them from the shortest student to the tallest student. You are allowed to do so with different techniques



a) Building a Sorted Line: Start from the right end of the line. Consider one student at a time and insert that student into the correct position among the students already arranged on the right, keeping that part of the line sorted.
1) Which sorting algorithm does this method resemble? (2 marks)
2) If you can instantly determine the correct position for each student, what is the overall time complexity? (1 mark)

b) Adjacent Swaps Only: You are only allowed to swap two neighboring students in the line.
3) Which sorting algorithm does this method resemble? (2 marks)
4) What is its time complexity? (1 mark)

c) Dividing and combining: You repeatedly divide the line into two smaller groups until each group contains only one student. Then, the teacher combines the groups back together by always placing the shorter student first.
5) Which sorting algorithm does this method resemble? (2 marks)
6) What is the time complexity of this algorithm? (1 mark)

d) Picking and moving a student: You are only allowed to pick a student from anywhere in the line and move the student to the correct position, shifting the other students as needed.
7) Which sorting algorithm does this method resemble? (2 marks)
8) What is the time complexity of this algorithm? (1 mark)

e) Building a Height Pyramid: Each time, the tallest student among the remaining students is identified and moved to the end of the line. This process continues until all students are in the correct order.
9) Which sorting algorithm does this method resemble? (2 marks)
10) What is the time complexity of this algorithm? (1 mark)

## PART 3: ALL QUESTIONS ARE REQUIRED [30 marks]



الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Republic of Egypt

CamScanner

## Question 4: (10 marks)

a) Find the complexity of the following blocks of code, where Arr is an array, n is the input size: [Note: your answer must show the steps that lead to your final answer] (1 mark each)

1)
```
val =1
n = 100
for k = 1 to n
        val = val × k
 end for
 for I = 1 to n
        A[I] = A[I] + val
 end for
```

2)
```
for i = 1 to 10 do
    for k = 1 to 10 do
        for (j = 2 ; j < n ; j *= 2)
            A[i] = i + k + j;
        end for
    end for
end for
```

3)
```
for i = 1 to n do
    for k = 1 to n do
        j = 1 ; m = n;
        while j <  m do
            m = m / 2;
        end while
    end for
end for
```

4)
```
for i = 1 to n Do
        A[i] = 1 / i
endfor
Heap_Sort(A, n)
for i = 1 to n Do
        B[i] = i²
endfor
Merge_Sort(B, n)
```

5)
```
i = 1, j =1
while i < n do
        i = 2 × i
endwhile
while j < n do
        j = 2 × j
endwhile
```

6)
```
def H(n):
  if n <= 1:
    return 1
  else:
    return H(n-1) + H(n-2) + H(n-3)
```

7)
```
def P(n):
  if n <= 1:
    return 1
  else:
    return P(n/2) + P(n/2)
```

8) The algorithm solves the problem of size n by recursively solving 4 sub-problems of size (n – 1) and then combining the solutions in constant time.

9)
```
def R(n):
  if n <= 1:
    return 1
  else:
    return R(n-1) + R(n-2) + 1
```

10)
```
def Q(n):
  if n <= 1:
    return 1
  else:
    return Q(n/2) + 1
```

## Question 5: (10 marks)

Mariam has a bag of candies. Each candy has a number written on it that shows how many points it is worth. She wants to divide the candies into two groups so that both groups have the same total number of points. Given a set of positive integers, determine whether the candies can be split into two groups with equal total points.

| Ex | input | output |
|---|---|---|
| 1 | S = {4, 5, 1, 2, 8} | Yes |
| 2 | S = {7,3, 2, 1} | No |

a) Design an efficient algorithm to do solve the problem (6 marks),
b) Analyse the complexity of your solution. (4 marks)

End of quest



الجامعة العربية المفتوحة-مصر
Arab Open University
القبول والتسجيل
الامتحانات

AOU EGYPT

CamScanner